

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 6

~ BookStream #002.01

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all right so now we are in chapter six a recipe for Designing active inference models Abraham Lincoln give me six hours to chop down a tree and I will spend the first four sharpening the ax all right Ali what does the quote mean and please lead us in to this discussion okay so as the opening quotation suggests it's about uh honing these skill sets uh and skill set for applying active inference framework to model uh the actual empirical uh situations that we want to we may use active inference for and it frames it as a four-step recipe to actually do the this kind of model uh although it's turned as recipe but uh I believe it's more like a guideline and it's not like I don't know cooking recipe that needs to be uh needs to be observed strictly so it's more like suggestions or guidelines to uh to begin to apply active inference framework in any kind of situation uh we'd like so uh section 6.2 outlines those four steps so the first step is which system are we modeling uh in terms of how we can Define the market blankets that most effectively and efficient the problem or question and we want to explore through that modeling through the modeling of that situation so uh first step is about defining those boundaries of a through marker blanket and the second step is about what's the most appropriate form for the generative model because obviously we can have many different types of uh generative models depending on again what phenomena or what questions we're trying to address the third stage is how to set up the generator model so after settling upon the type of degenerative model we want to use we can talk about the details of the generator model and how it can be efficiently modeled to be both tractable and also useful to address the situation of interest and finally how to set up the generative process so as we saw in chapter two uh there's a important uh that slight distinction between generative process and generative model so they're not always the same but they're closely related to each other generates a process to the hidden states of the external world and generative model is how the agent infer about those hidden States provided by generative process so uh considering that the

environment is always very complex to be modeled precisely and exactly with every parameter accounted for we need to have a restricted set of parameters to account for in the generative process so that's what stage four deals for deals about so all of these steps even if not uh observed or followed sequentially are really I mean essential to setting up I mean or the initial setup of uh the situation we're trying to model and uh I cannot imagine how any scenario can progress without observing any of these any of these steps at least uh I mean just by thinking about how to set up the situation or scenario awesome and in the rest of the rest of the chapter yes just wanted to um highlight again these steps don't have to be followed in order they really are like a summary as Mark has just written in the live chat here it really is like a summary of the earlier parts of the book in terms of things that you want to capture in your consideration and also it's a great way to connect a lot of the technical ideas that are brought up about generative model generative process and so on with some bigger questions like who are we why are we making this model how is the model going to be used so we take that in a lot of different ways in the textbook groups but which system are we modeling what is the most appropriate form for the generative model how should we set up the generative model and how to set up the generative process are all questions that must be addressed at least when actually coming through with carrying out a theoretical or empirical octave inference modeling task and the following sections are going to go into more detail on each of those four questions so Ali do you want to go for this 6'3 and carry on okay so section 6613 is the elaboration of the first step that we just saw which I mean what systems are we modeling and by this question we mean how I mean what kind of system uh I mean how the system carves up the state space using the markup blanket or more precisely the active States and sensory States and I believe it is and both one of the most fundamental and also one of the probably one of the most challenging steps uh to be carried out because um well markup blanket is just a conceptual tool we use to frame uh the natural phenomena or any other scenario we want to model so that scenario or that model obviously doesn't care about how we set up our markup blanket or how we need our markup blanket but uh deciding on a proper a set of active States and sensory states that would allow us to address the problems we need to explore or the questions we need to answer is not a trivial task and it needs lots of lots of consideration and even creativity on part of the researcher or anyone who wants to design an active inference based model for the problem so that's where the I mean the significance of doing doing these kinds of modeling with the insights and creativity of a a researcher is I mean will be obvious because it's not just something mechanical or uh despite the fact that the word recipe uh can

somehow make us believe that it's just a recipe to follow but uh in actuality it's not the case at all so that's the discussion of Marco blanket and how what criteria we need to keep in mind to settle upon the proper uh yeah proper markup blanket around our system and the next section what is the most appropriate for form for the generative model so here again we can have various way to set up our generative model in terms of whether it's discrete or continuous variables we want to account for about the time scale is it shallow or hierarchical about uh or about the temporal depth of the inference and planning so these are all problems we need to be precise about or questions we need to precise about and whenever we want to set up our generative model and then we go to section 6.5 how to set a sorry yeah how to set up the generator model so after the precisely describing what needs to be accounted or what parameters and the nature of the generative model needs to be accounted for uh then we will probably need to address this question of how can that generative model be actually set up so setting up the variables for the generative model will I mean we will be able to be concerned about how those matrices as a b c and d matrices matrices can be defined as we saw in chapter four and then which parts of the generator model are fixed and what is learned so again it's really important to know what are the Dynamics of the of those parameters are in terms of the learning process and what what needs to be updated consistently or learned as opposed to the variables that uh either don't require consistent updating or they vary quite slowly as compared to the time scale of the generative model and then we go through the process of setting up the generative process and as I mentioned earlier it is probably one of the most complicated uh steps of the recipe because of the complexity the I mean the enormous complexity of any real-time situation and particularly the environment that the agent uh will act and behave but even if not having uh I mean even if there is not any um strict or I don't know obvious way to go through the through this stage of setting up the generative process this section provides some helpful criteria or helpful points in order to help us in that stage as well so finally after going through all of these stages we need to somehow get I mean obviously we need results and we need to provide our results in a we get our results from our active inference model and uh section 6.7 addresses that uh I mean the the that very idea of how to simulate visualize analyze and fit data using active imprints because it's not just enough to set up the model we need to get the model to work so how can we feed the data into the model how can we read the data and finally how can we analyze or analyze the simulated data through the active inference so this is uh in a way um a kind of preliminary recipe or preliminary procedure for doing active inference or an active inference modeling or at least suggested way uh to uh to do those kinds

of modeling but in the subsequent chapters in particular chapters seven and eight will go through some uh helpful cases studies uh for how to actually use these procedures um to model some real world situations awesome thanks well there's a lot to say on chapter six one piece that comes up a lot is what is in the active inference kernel the cognitive kernel and what are situations that we can model with active inference there's a huge variety of cognitive phenomena and statistical outcomes that we might be interested to study like counterfactuals and planning like multi-scale attention and covert action like the complex ways in which memories might um influence decision making in the moment based upon associations made between different sensory modalities every amazing real world situation you can imagine and it's really helpful to think about active inference as more like a framework of interoperable motifs that we can compose and indeed even be creative with rather than giving us all the answers at the core model because the kind of essential minimal active inference model doesn't necessarily include every single attribute that you might be interested in just like any given linear regression isn't going to include every single feature that you're looking for this is happening in a different type of modeling framework but still there's a lot of customization and development that goes into adapting or elaborating an active inference type model for your given system or scenario of Interest chapter 6 gives us some helpful guidelines and things to consider which we've unpacked and expanded on elsewhere and in the textbook group we work together to characterize different systems of interest to map different parts or different observables from that system using the active inference ontology into a unified active inference type model so chapter 6 starts the second half of the book which is the more practice oriented part of the book and the rest of the part two is going to be a lot more about the specific kinds of generative models that you can use discrete and continuous time and about using them with empirical data in chapter nine but chapter six just stands alone as a summary of part one of the book in some ways and a set of four questions which system or remodeling the generative model how to set up the generative model and how to set up the generative process so that we can be active inference model ourselves all right well Ollie thanks so much for all of these great uh conversations and cohorts we will be able to process this video to clip out these first versions of the background and context overviews and we'll distribute them to the cohorts so that people can show up with their Curiosities and passions and we'll see where that goes any last thoughts thank you it was a pleasure to go through these chapters once again with each time obviously with uh a bit more understanding than than the previous one so I'm looking forward to the next awesome all right thanks Ali farewell